

Soft Robotic Proprioceptive Posture and Positioning Enhancement via Distributed Multifield Magnetic Inductive Sensing

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Abstract—Soft robots’ highly flexible and adaptive movements drive demand for sensing methods to control their posture and position. Recently, magnetic sensing has shown promising perspectives in advancing soft robotic applications, offering high design adaptability and exceptional sensing performance by eliminating the need for soft-material substrates. In this study, we further explore the utilization of magnetic inductive sensing mechanisms with multiple magnetic fields in soft pneumatic robots to track distributed key points, enabling flexible and versatile configurations of soft robotic modules over a large sensing range. We introduce a Soft Robotic Distributed Multifield Magnetic Inductive Sensing (SoRo-MIS) approach that configures magnetic coils at key distributed points within the soft robotic bodies to obtain proprioceptive posture information and extend the sensing range by overlapping multiple magnetic fields. It facilitates a modular design capable of assembling and disassembling different soft robotic modules without compromising the overall system or requiring a redesign, expanding the sensing functionality from proprioceptive posture to mutual positioning, and demonstrating the modularity and scalability of the system. The methodology, from concept, modeling, design, fabrication, to validation, enables motion feedback control in pneumatic soft robotic systems. We demonstrate the potential for expanded applications in a soft two-segmented robot, a soft finger-assembled arm, and a two-crawling robot system, showcasing its high flexibility in modular design, as well as high sensing performance in large ranges for robotic motion control.

Index Terms—Soft robotics, soft sensor, proprioceptive posture, position sensing, magnetic, multifield, inductive sensing.

I. INTRODUCTION

SOFT robotics has rapidly grown in recent years. Unlike traditional rigid robots, soft robots utilize soft materials to generate large and continuous deformation, enabling seamless adaptation to various situations. This makes them ideal for applications like manipulation, exploration, and navigation [1]-[6]. Within these applications, proprioceptive posture sensing and spatial positioning play a crucial role in feedback control and decision-making. However, the large and continuous deformations occurring within individual robots, combined with the physical disconnection among multiple robots, pose substantial challenges in obtaining accurate sensory information [7],[8].

Over the past few decades, various sensing approaches for soft robots have been extensively explored [9]-[23]. Some approaches involve the development of soft strain sensors [9]-[16] that can be locally integrated into the robot’s body using large-strain substrates, along with optical, capacitive, or resistive sensing mechanisms. This enables the robot to collect rich sensory information, such as changes in its shape, pressure, or force based on dedicated analytical or data-driven models [10],[14]. More recently, non-strain-based sensing methods have been developed, utilizing optical and magnetic approaches to locate the distal points instead of conventional strain deformations to retrieve the sensing information [17]-[24] (see Table 1). Among them, magnetic sensing can sense through obstacles without limitations of line-of-sight, offering flexibility in sensor placement. They demonstrate high dynamic performance and precision in capturing robotic movements within a single entity [18],[21]. Our previous work on Hall effect-based magnetic sensing employs three hall effect chips arranged around a permanent magnet, as illustrated in Fig. 1(a), to retrieve the position and orientation of the distal plate in soft robots [18]. However, its effective sensing range was restricted by the strength of the magnetic field from the permanent magnet. Instead of detecting magnetic strength directly, magnetic inductive sensing measures change in magnetic flux where transmitting and sensing coils are flexibly configured at different frequencies [17],[23]. As a result, this approach provides an alternative to expand the sensing range and offers high design adaptability by configuring multiple magnetic fields. Recent advancements in magnetic inductive sensing utilizing several transmitting coils and one sensing coil showing superiority on large surface contact points and force detection [22],[23]. Additionally, overlapping multiple magnetic fields at different frequencies has demonstrated significant capabilities for large-scale multidimensional posture sensing in applications, as seen in commercial motion tracking systems [25], rigid surgical robots and cable-driven continuum robots, highlighting the potential of magnetic inductive sensing in robotic information retrieval [26]-[28].

In this study, we further explore the utilization of a magnetic inductive sensing mechanism with multiple magnetic fields in soft pneumatic robots for proprioceptive posture and position

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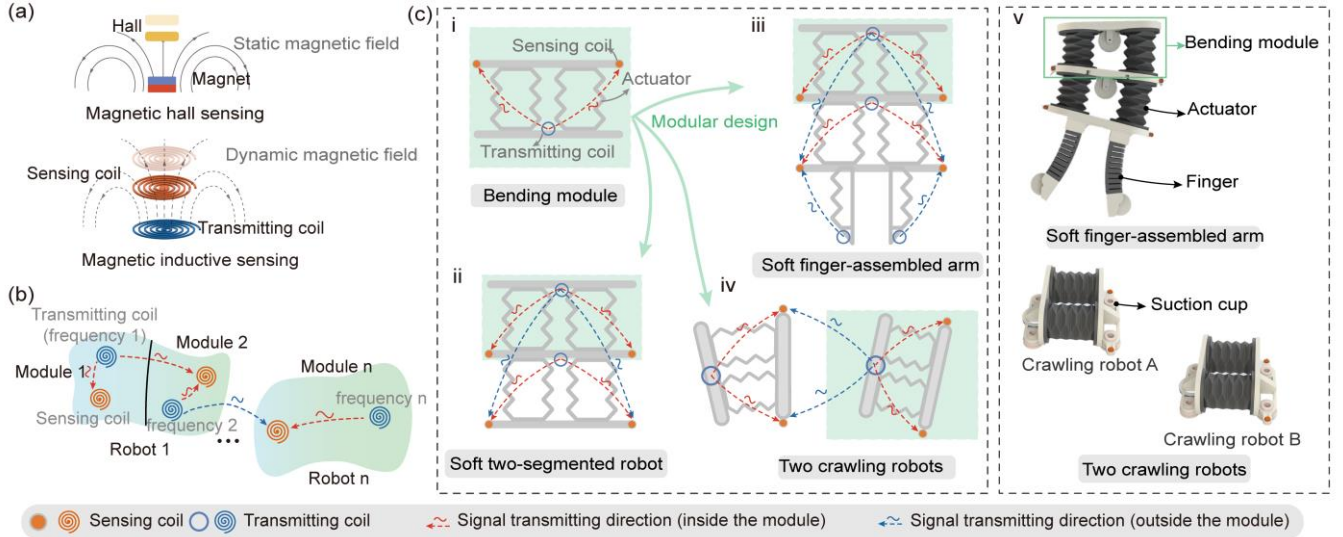


Fig. 1. Overview of Soft Robotic Multifield Magnetic Inductive Sensing (SoRo-MIS) concept. (a) Related magnetic sensing methods. (b) Schematic diagram of the SoRo-MIS concept that enables adaptable coils configurations and an expandable sensing range within multiple magnetic fields for both soft robotic proprioception and mutual position sensing. (c) Concept validation in soft robotic systems: (i) Bending module. (ii) Soft two-segmented robot. (iii) Soft finger-assembled arm. (iv) Two crawling robots. (v) Prototypes of soft robots.

TABLE I. SENSING PERFORMANCE COMPARISONS TO THE STATE-OF-THE-ART IN SOFT ROBOTICS

State of the art	Sensing methods	Sensing range(mm)	Sensing accuracy(mm)	Sensor size(mm)	Sensing modality			Modularity	
					Proprioceptive posture	Mutual position	Force	Module assembly/disassembly	Sensing methodology
Sundaram [21] 2023	Static magnetic field and magnetometer	68(L)	0.1	50 (L)×50(W)×5(H)	✓	NA	NA	NA	Fixed pair
Soleimani [22] 2023	Mutual Inductance	400	1	40 (D)×8(H)	✓	NA	✓	NA	Optional transmitter-sensor pair
Yi [18] 2025	Hall Effect	60(L)	0.36	4(L)×3(W)×1.5(H)	✓	NA	✓	NA	Fixed pair
Yang [24] 2019	Hall Effect	80(L)	0.4	4(L)×3(W)×1.5(H)	✓	NA	NA	NA	Fixed pair
Bono [16] 2024	Optical waveguides	40(L)	2	0.6(D)×20(H)	✓	NA	NA	NA	Fixed pair
This work	Multifield magnetic inductive	270	0.31	6(D)×8(H)	✓	✓	NA	✓	Optional transmitter-sensor pair

Note: L is the abbreviation for length, D for diameter, W for width, and H for height. Modularity refers to the ability to assemble and disassemble different soft robotic modules without compromising the overall system or requiring a redesign. NA means not applicable.

sensing, while leveraging flexible and versatile configurations of soft robotic modules over an expandable sensing range (listed in Table 1). We introduce an approach of Soft Robotic Distributed Multifield Magnetic Inductive Sensing (SoRo-MIS) that embeds coils within multiple magnetic fields at key distributed points in the pneumatic soft robotic bodies to obtain proprioceptive posture information. These key points are defined based on the desired position and orientation of the distal plates in each module, leveraging the widely accepted constant curvature assumption to model the bending behavior of soft robots. Magnetic coils are directly embedded at these points without the need for physical connections or soft material substrates, forming sensor-integrated soft robotic modules in which each sensing coil can detect signals from multiple transmitting coils. This distinguishes our approach from traditional magnetic and optical waveguide sensing methods. They only measure local deformations instead of capturing key distributed points to reconstruct full-body posture, and therefore rely strictly on fixed transmitter-sensor pairs with

pairs not interconnected from a sensing perspective. These modules can be assembled and disassembled without compromising the overall system or requiring a redesign, further expanding the sensing functionality from proprioceptive posture to mutual positioning, and demonstrate the modularity and scalability of the system (Fig. 1(b)). These capabilities are validated in a soft two-segmented robot, a soft finger-assembled arm, and a two-crawling robot system (Fig. 1(c)).

The main contributions of this work include, 1) approach of SoRo-MIS with principles of defining proprioceptive posture of soft pneumatic robotic module through key distributed points and configuring magnetic coils into these points to extend the sensing range by overlapping multiple magnetic fields, 2) modular design method capable of assembling and disassembling different soft pneumatic robotic modules without compromising the overall system or requiring a redesign, expanding the sensing functionality from proprioceptive posture to mutual positioning, and demonstrating the modularity and scalability of the system, 3) the utilization of

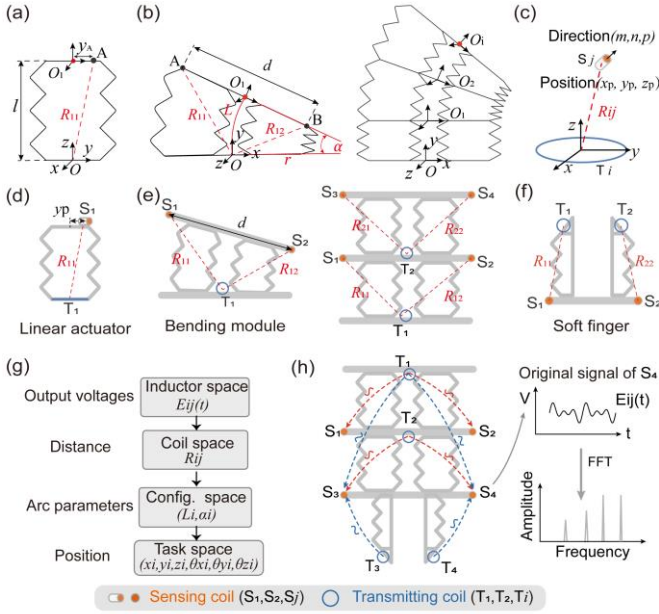


Fig. 2. Modeling of SoRo-MIS for soft robots. (a) Schematic of modeling of linear motion and (b) bending motion. (c) Magnetic dipole model. (d)(e)(f) Schematic of coils configuration of linear actuator, bending module, and soft gripper. (g) Kinematic mapping. (h) The sensing coil signal is processed using FFT to extract the amplitude corresponding to the transmitting frequency.

sensory information for target posture control and mutual motion following of the modular soft robotic systems, demonstrated in a soft two-segmented robot, a soft finger-assembled arm, and a two-crawling robot system, with a comprehensive methodology including sensing modeling, design, fabrication, experiments, and soft robotic demonstrations.

This paper is organized as follows: Section II introduces the SoRo-MIS concept, explained with the modeling of soft robots' proprioceptive posture and positions, as well as the modular design. Section III presents the preliminary validation of the core concept and the fabrication process of the proposed soft robotic modules. It evaluates the design's performance in measuring both linear and bending motions. In Section IV, we demonstrate and showcase its potential application in a soft two-segmented robot, a soft finger-assembled arm, and a multi-robot system.

II. CONCEPT AND MODELING

A. Concept

The SoRo-MIS approach is built on the principle of defining the proprioceptive posture of soft robotic modules through key distributed points, where magnetic coils are strategically embedded. This framework allows seamless assembly and disassembly of different soft pneumatic robotic modules without compromising system integrity or requiring redesign, while expanding the sensing functionality from proprioceptive posture to mutual positioning.

The core concept is detailed in this section as the following parts: 1) soft robotic proprioceptive posture modeling part elaborates the key points to define the posture of soft robotic

with geometrical analysis on linear and bending motions; 2) magnetic coils configuration for soft robotic module part describes the arrangement and positioning of transmitting and sensing coils for various soft modules.

B. Soft Robotic Proprioceptive Posture Modeling

Following the widely adopted Constant Curvature assumption [29]-[31], which models the bending of a module as a segment of a circle with a uniform radius of curvature, both linear and bending motions can be fully defined by the position and orientation of the center point O_i on the rigid distal plate, which is

$$\xi_{O_i} = [x_i \ y_i \ z_i \ \theta_{x_i} \ \theta_{y_i} \ \theta_{z_i}]^T. \quad (1)$$

Regarding the soft linear actuator moving along the z -axis in Fig. 2(a), its proprioceptive posture can be described by

$$\xi_{O_1} = [0 \ 0 \ z_1 \ 0 \ 0 \ 0]^T. \quad (2)$$

Based on geometric relations, the actuator's posture can be represented as,

$$z_1 = \sqrt{R_{11}^2 - y_A^2}, \quad (3)$$

where R_{11} is the distance from any point A ($0, y_A, z_A$) on the distal plate to the center of the bottom plate. Equation (3) indicates that the proprioceptive posture of the linear soft actuator ξ_{O_1} , i.e., z_1 , is determined by R_{11} .

Multiple parallelly arranged linear soft actuators form a soft robotic bending module, enabling linear contraction, extension, and bending motions in space. These motions exhibit rotational symmetry. Thus, for illustration, we take a representative case in a two-dimensional plane with two actuators for a module, as shown in Fig. 2(b). In this module, the position and orientation of the central point are defined as

$$\xi_{O_1} = [x_1 \ y_1 \ 0 \ 0 \ 0 \ \theta_{z_1}]^T, \quad (4)$$

which is simplified to $\xi_{O_1} = [x_1, y_1, \theta_{z_1}]^T$. According to the Constant Curvature assumption and geometric relations, we have

$$\begin{bmatrix} x_i \\ y_i \\ \theta_{z_i} \end{bmatrix} = \begin{bmatrix} (c_i + d/2)(1 - \frac{(c_i + d/2)^2 + c_i^2 - R_{ij,j=2}^2}{2c_i(c_i + d/2)}) \\ (c_i + d/2)\sin(\arccos(\frac{(c_i + d/2)^2 + c_i^2 - R_{ij,j=2}^2}{2c_i(c_i + d/2)})) \\ \arccos(\frac{(c_i + d/2)^2 + c_i^2 - R_{ij,j=2}^2}{2c_i(c_i + d/2)}) \end{bmatrix}, \quad (5)$$

where $c_i = (d^2/4 - dR_{ij,j=2}^2)/(R_{ij,j=2}^2 - R_{ij,j=1}^2)$, d is the distance between points A and B in the distal plane. The singularity of this kinematic model arises when $R_{11} = R_{12}$. In this case, the module exhibits linear motion, following the model described in Equation (3). Equation (5) indicates that the proprioceptive posture of the soft robotic bending module depends on the distances R_{ij} . Deviations from Equation (5) are provided in the Supplementary Materials.

Bending modules can be longitudinally stacked to form a soft multi-segmented robot. Equation (5) obtains the posture of each segment. The end position of the robot can be derived using the homogeneous transformation matrices. We denote the position

component $[x_i, y_i, z_i]^T$ of the posture of the i_{th} segment as $\mathbf{p}^{oi}$. Using the coordinate $\{O\}$ in the 1_{st} segment as the base frame, $\mathbf{p}_0^{oi}$ denotes the representation of $\mathbf{p}^{oi}$ in the base frame

$$\begin{bmatrix} \mathbf{p}_0^{oi} \\ 1 \end{bmatrix} = \mathbf{T}_1^2 \mathbf{T}_2^3 \cdots \mathbf{T}_i^{i+1} \begin{bmatrix} \mathbf{p}^{oi} \\ 1 \end{bmatrix}, \quad (6)$$

where $\mathbf{T}_i^{i+1}$ is the homogeneous transformation matrix of the coordinate system $\{O_i\}$ in $(i+1)_{th}$ segment relative to the coordinate system $\{O_{i-1}\}$ in the i_{th} segment, and is detailed in the Supplementary Materials. Equation (6) is the model for the robot's end position, and orientation can be achieved by incorporating Equation (5) into (6). In summary, the proprioceptive posture, encompassing both linear and bending motions, can be defined through distributed points on the distal plate and determined by geometric distances.

C. Magnetic Coils Configuration for Soft Robotic Module

These distances can be measured by pairs of transmitting and sensing coils with principles of configuring their placements in terms of location and direction. Fig.2 (c) illustrates the location and orientation of the sensing coil in the coordinate system centered at the transmitting coil, defined by its position $\mathbf{P} = (x_p, y_p, z_p)$ and normal vector $\mathbf{H} = (m, n, p)$, $m^2 + n^2 + p^2 = 1$. A sinusoidal signal is applied to the transmitting coil. According to Faraday's law and Biot-Savart law [32], a corresponding induced electromotive force is generated in the sensing coil. The frequency of the induced signal is the same as that of the transmitting coil, and its amplitude depends on the relative distance and orientation between the coils. The induced signal of the sensing coil $E_{ij}(t)$ can be formulated as:

$$E_{ij}(t) = k \left(\frac{3(\mathbf{H} \cdot \mathbf{P})\mathbf{P}}{R_{ij}^5} - \frac{\mathbf{H}}{R_{ij}^3} \right) \mathbf{S} \quad (7)$$

where $R_{ij} = (x_p^2 + y_p^2 + z_p^2)^{1/2}$ is the distance between two coils; $k = \mu N_s N_t I_0 \omega \cos(\omega t) r_t^2 / 4$, μ is the magnetic permeability; N_s and N_t are the turns of the sensing and transmitting coils; $I = I_0 \sin(\omega t)$ is the excitation current, $\omega = 2\pi f$, and r_t is the radius of the transmitting coil. Deviations from Equation (7) are provided in the Supplementary Materials. Equation (7) reveals that the signal of the sensing coil depends on the hardware setup of the transmitting coil, as well as the location and direction of the sensing coil. The hardware setup of the transmitting coil will be detailed in the Fabrication section. Regarding the location and direction of the sensing coil, the coils can be placed in a specific direction so that their relative orientation remains unchanged during the motion of the soft robot, ensuring that $\mathbf{H}$ is constant. Equation (7) is thus simplified so that position is the only remaining variable. Therefore, for a linear soft actuator, which generates linear motions without directional changes, the transmitting and sensing coils can be independently positioned on the fixed bottom plate and the distal plate, respectively, as shown in Fig. 2(d). With this configuration, $\mathbf{P} = (0, y_p, z_p)$ and $\mathbf{H} = (0, 1, 0)$. Substituting them into Equation (7), we have

$$E_{ij}(t) = N_s k \left(\frac{3y_p \sqrt{R_{ij}^2 - y_p^2}}{R_{ij}^5} \right), \quad (8)$$

where y_p is a constant representing the fixed position of point P on the distal plate. Therefore, the voltage of the sensing coil

varies only with the distance R_{ij} . For a soft robotic bending module, the transmitting and sensing coils are configured with the same normal vector (Fig. 2(e)) to simplify calculations. Furthermore, both are positioned within the movement plane to maintain a constant normal vector during the motions, i.e., $\mathbf{P} = (x_p, y_p, 0)$ and $\mathbf{H} = (0, 0, 1)$ for Equation (8),

$$E_{ij}(t) = N_s k \left(\frac{1}{R_{ij}^3} \right) \quad (9)$$

where the voltage of the sensing coil also varies only with the distance R_{ij} . Following this principle, the soft finger in Fig. 2(f) adopts the same coil configuration as the soft bending module, as its movements also follow a constant-curvature arc.

When the soft modules are assembled, the sinusoidal signals applied to the transmitting coils in each module are set to different frequencies $f_1, f_2, \dots, f_i$. According to the linearity of Maxwell's equations, the resultant magnetic field at any point in space generated by multiple transmitting coils is simply the vector superposition of the individual magnetic fields. In this case, the signal induced in sensing coil j is equivalent to the superposition of the contributions from all transmitting coils acting individually, i.e., $E_j(t) = \sum_i E_{ij}(t)$. By applying Fast Fourier Transformation (FFT) to convert the composite signal into the frequency domain, we can obtain the corresponding amplitude at each frequency that enables the independent measurement of distances between the sensing and transmitting coils, as shown in Fig.2 (h).

D. Signal Decoding Process

The signal decoding process transforms voltages $E_{ij}(t)$ in the inductor space into $\xi_{O_i}(x_i, y_i, z_i, \theta_{x_i}, \theta_{y_i}, \theta_{z_i})$ in task space, as shown in Fig.2(g). Although Equation (8)(9) establishes the relationship between output voltages and distances under a specific hardware setup, the output voltages are still influenced by nonlinearities in magnetic fields and environmental interferences, which prevent the model from accurately decoding magnetic signals without significant empirical adjustments. To address these challenges, calibration with fitting functions is commonly used to correlate output voltages and distances in current research [22]. The relationship between the output voltages of the sensing coil and distances can be experimentally derived using fitting functions $f_{R_{ij}}$, which follow

$$R_{ij} = f_{R_{ij}}(E_{ij}), \quad (10)$$

where $f_{R_{ij}}(E_{ij}) = e_1 E_{ij}^{e_2} + e_3$, E_{ij} is the amplitude of $E_{ij}(t)$. Parameters e_1 , e_2 , and e_3 are constants derived from function fitting. Substituting (10) into (5), we have the kinematic model for the i_{th} segment,

$$\begin{bmatrix} x_i \\ y_i \\ \theta_{z_i} \end{bmatrix} = \begin{bmatrix} (c_i + d/2) \left(1 - \frac{(c_i + d/2)^2 + c^2 - f_{R_{ij}}(E_{ij})^2}{2c_i(c_i + d/2)} \right) \\ (c_i + d/2) \sin(\arccos(\frac{(c_i + d/2)^2 + c_i^2 - f_{R_{ij}}(E_{ij})^2}{2c_i(c_i + d/2)})) \\ \arccos(\frac{(c_i + d/2)^2 + c_i^2 - f_{R_{ij}}(E_{ij})^2}{2c_i(c_i + d/2)}) \end{bmatrix}, \quad (11)$$

where $c_i = (d^3/4 - df_{R_{ij,j=2}}(E_{ij,j=2})^2) / (f_{R_{ij,j=2}}(E_{ij,j=2})^2 - f_{R_{ij,j=1}}(E_{ij,j=2})^2)$. In Equation (11), multiple sets of distances involve multiple induction signals. The induced electromotive force amplitude

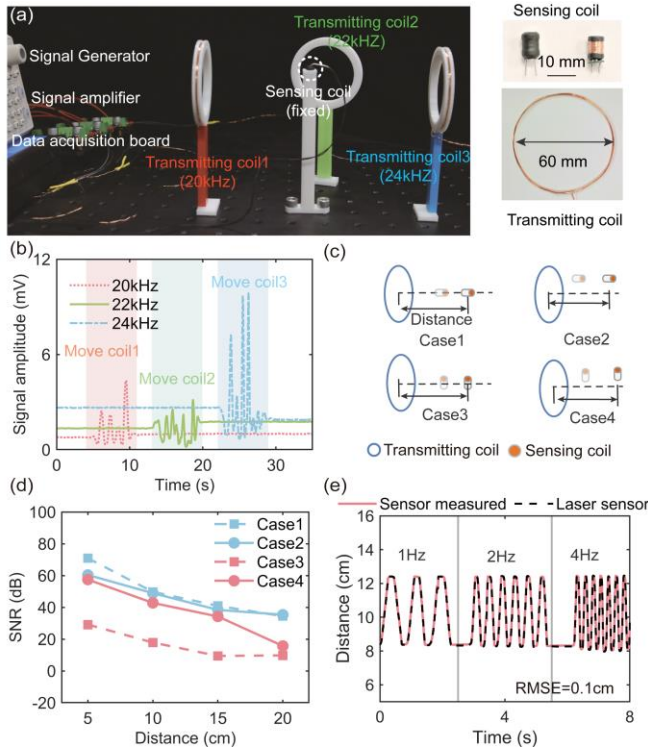


Fig. 3. Concept validation. (a) Experimental setup for validating the proposed concept. (b) The amplitude variation decomposed from the original signals at three different transmitting frequencies. (c)(d) Signal-to-Noise Ratio of the sensing coil in four cases. (e) Sensor's response at various movement frequencies: 1 Hz, 2 Hz, 4 Hz in Case 3.

E_{ij} for each coil pair is extracted from the composite signal $E_j(t)$ using FFT, enabling independent measurement of the distances between the sensing and transmitting coils. This ensures that the signal decoding process within each module does not interfere with others. When the modules are assembled, the pose of each module can still be determined using Equation (11).

III. CONCEPT VALIDATION IN LINEAR MOTION

A. SoRo-MIS Hardware System Validation

The hardware system comprises transmitting and sensing coils, a signal generator, a constant current regulator, signal acquisition hardware, and a computer for data processing. The transmitting coils were fabricated by winding 0.3 mm copper wire into circular planar coils on 3D-printed ring-shaped bases, with diameters adjustable to suit different robotic dimensions. The sensing coil employed a 10mH I-shaped magnetic core inductor (Diameter: 6mm, Length: 8mm). The signal generator (UTG962, UNI-T) and the constant current regulator (OPA57, Texas Instruments) were used to drive the transmitting coils. The output signal of the sensing coil was acquired using a synchronous digital acquisition card (USB2872C, ART-Technology, 8-channel, 250 kHz) and transmitted to the PC for processing with FFT in Python. By dividing the sampling frequency 250Hz by the number of samples 3000, we obtained that the frequency interval of the current for FFT must not be lower than 83 Hz. A larger frequency interval enhances robustness to noise and improves the identification of prominent frequency components. We selected a frequency of

approximately 20 kHz and a minimum frequency interval of 1 kHz for the signal tests of multiple magnetic fields.

During the tests, the output signals were recorded while three transmitting coils were sequentially moved around a single sensing coil, as illustrated in Fig. 3(a). The effective value of the current was set at 300 mA due to hardware system capacity limitations. The signal frequencies of the coils were set to 20 kHz, 22 kHz, and 24 kHz, respectively. Fig. 3(b) shows that the output voltage amplitudes respond independently to the individual motion of the transmitting coils. This demonstrates the sensing coil's ability to detect positional changes of the transmitting coils across different magnetic fields.

The signal-to-noise ratio (SNR) was further evaluated on this experimental platform. We measured the SNR under four conditions, placing the sensing coil either at the center or off-center of the transmitting coil, with varying orientations, as shown in Fig. 3(c). The results, shown in Fig. 3(d), indicate that Cases 1, 2 and 4 present SNR over 20 dB within a 200 mm distance. These results demonstrate that the position and orientation of the sensing coil within the 200 mm sensing range maintain an acceptable error tolerance for distance sensing in soft robots without compromising sensing accuracy. In contrast, Case 3 exhibits a relatively lower SNR of 9.8 dB due to the orthogonal placement at the center. Therefore, Case 3 should be avoided in the following robotic design.

Taking Case 3 as an example, we conducted a preliminary test to evaluate the sensor's response at various movement frequencies. Following the setup in Case 3, the sensor was moved rapidly at frequencies of 1 Hz, 2 Hz, and 4 Hz with an amplitude of 40mm, using a laser sensor as the ground truth. The results in Fig. 3(e) show no significant time misalignment between the sensor signal and the ground truth, indicating that the FFT does not introduce any substantial delay to the sensor system.

B. Fabrication of SoRo-MIS Actuator

The fabrication of SoRo-MIS actuator proceeds in four steps: a) fabricating the origami chamber (Rubber Hei-Cast 8400, shore hardness 80A) by the injection molding process, design details on the origami chamber could be found in our previous work [33]-[35]; b) inserting the transmitting and sensing coils into the 3D-printed bottom plate and distal plate (FDM printer, PLA materials); c) assembling these two plates into two ends of the chamber; d) embedding the transmitting and sensing coils follows the position and orientation configurations detailed in the above section. In the proposed actuator prototype, as shown in Fig. 4(a), the dimensions of the chambers and plates are the same with a diameter of 66 mm. The overall height of the actuator is 90 mm.

C. Concept Validation in Linear Actuator

Two SoRo-MIS linear actuators were fabricated to preliminarily validate their capability for measuring proprioceptive posture and overlapping two magnetic fields to expand the sensing range. Fig. 4(b) illustrates the signal processing flowchart of the linear actuator, including DAQ (data acquisition), FFT computation, controller computation,

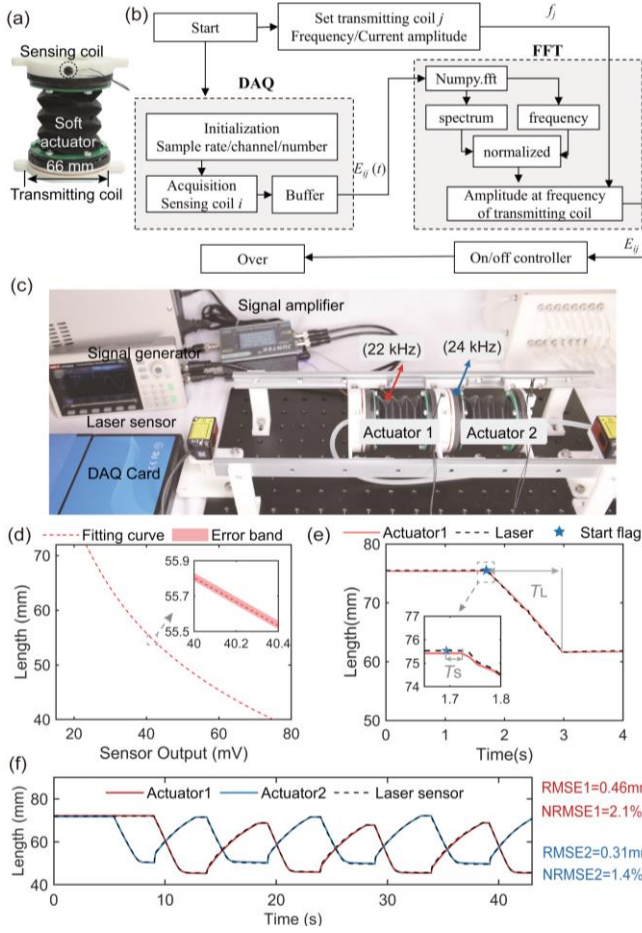


Fig. 4. Proprioception in linear motion. (a) Prototype of SoRo-MIS actuator. (b) Sensor signal processing flowchart. (c) Experimental setup. (d) Characterization of relationships between actuator's length and output voltage signal. (e) Time response of linear actuator. (f) Performance validation of simultaneously linear motions of two actuators within two

and control signal transmission to the actuator. An experimental platform (Fig. 4(c)) was constructed to actuate the actuators pneumatically and obtain the ground truth. Two laser sensors (Panasonic, HG-C1100, $50\text{mm} \pm 35\text{mm}$) were used to record the length of both actuators. The transmitting coils of two actuators were separately supplied with $f=22\text{ kHz}$, 24 kHz ; $I_0=300\text{ mA}$.

Firstly, we calibrated the relationship between the actuator length and the output signal of the sensing coil. Each actuator was calibrated five times, and the average curve was fitted. Fig. 4(d) presents the results for actuator 1. The maximum deviation within the error band is only 0.066 mV , accounting for 0.21% of the full signal range, demonstrating the high repeatability of the calibration results. Following the fitting function of Equation (10), we have $e_1=277.97$, 227.34 , $e_2=-0.244$, -0.280 , and $e_3=-57.05$, -40.98 , respectively, set for two SoRo-MIS actuators. A detailed calibration process can be found in the Supplementary Materials.

Furthermore, to investigate the delay introduced by the sensing mechanism, a step-response test was conducted on the linear actuator to record the sensor processing time. Among them, the DAQ acquisition window and FFT computation are the main sources of sensor delay. During the test, a flag was set at program initiation, with timers inserted to record the time

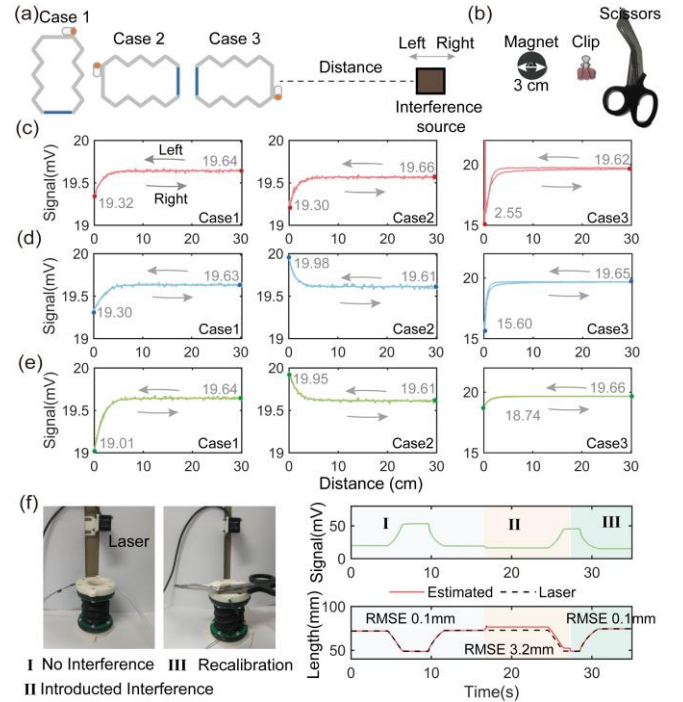


Fig. 5. Ferromagnetic interference test. (a) Overview of the experimental platform. (b) Images of the three interference sources. (c)(d)(e) Signal variation in the magnet, clip and scissors experiment. (f) Interference compensation through recalibration.

from initiation to actuator response (Fig. 4(e)). The total loop time T_L was 1285ms , while the sensing process time T_S was measured as 45ms , which accounts for only 3.5% of the entire loop time. Specifically, the average time consumed for data acquisition was 17ms and the FFT computation required an average of 0.99ms . This delay could be accepted for soft pneumatic actuation systems.

Then we conducted an experiment to measure the simultaneous motion of the two actuators to validate the capability of overlapping two magnetic fields to expand the sensing range. In this test, the middle plate remained fixed. The results in Fig. 4(f) demonstrate that the linear movements of both actuators can be independently and simultaneously measured. Each actuator's linear distance is over 20mm , resulting in a combined range of over 40mm . The sensing range can be further extended by incorporating additional magnetic fields. Furthermore, both actuators demonstrate high accuracy in length measurements, with Root Mean Squared Errors (RMSE) of 0.46mm and 0.31mm . The Normalized Root-Mean-Square Error (NRMSE) are 2.1% and 1.4% , respectively.

Ferromagnetic objects distort the spatial magnetic field and consequently affect the output signal of the sensing coils. To evaluate this effect, three interference sources (a magnet, a metal clip, and scissors) were tested under three actuator configurations (Cases 1–3), as shown in Fig. 5(a)(b). The output voltage was recorded as a function of the distance between the interference source and the actuator.

As shown in Fig. 5(c)(d)(e), the actuator remains insensitive to interference beyond 5 cm , with a maximum fluctuation of only 0.1 mV . When the distance is less than 5 cm , the signal

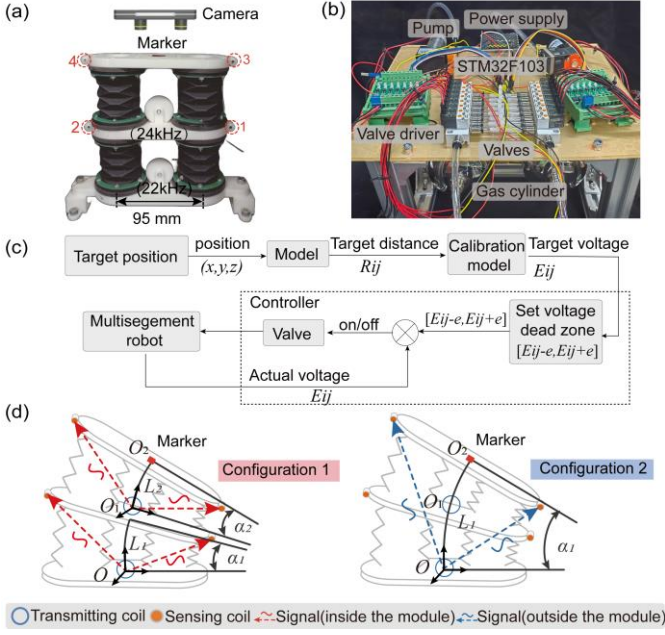


Fig. 6 Multisegmented robot prototypes and configurations. (a) Prototype of a soft two-segmented robot with an arrangement of the SoRo-MIS array. (b) Actuation hardware system. (c) Control strategy for the segmented robot. (d) Two coil configurations for distal position estimation.

variation increases sharply. In Case 1 and Case 2, the maximum signal variation decreases from 19.64 mV to 19.01 mV, corresponding to a predicted length error of 1.1 mm, which accounts for only 1.5% of the actuator length. In contrast, in Case 3, the maximum magnetic interference causes the induced signal to drop from 19.62 mV to 2.55 mV, resulting in a predicted length error of 86.8 mm, corresponding to 120.5% of the actuator length. Moreover, a noticeable recovery delay is observed in Case 3 when the interference source moves away, which is attributed to the magnetization of the ferromagnetic core inside the sensing coil. Therefore, external ferromagnetic objects should be kept at a sufficient distance from the sensing coil. For stable interference conditions, measurement errors can be compensated through recalibration. As shown in Fig.5(f), the RMSE is reduced from 3.2 mm to 0.1 mm after recalibration.

IV. EXPERIMENTAL VALIDATION ON MODULAR DESIGN

In addition to validating the linear actuators, this section explores the potential of modular design. Two soft robotic bending modules could be either stacked to form a two-segmented arm or separated into two crawling robots, enabling the measurement of both proprioceptive posture and mutual positioning over a large sensing range. This two-segmented arm has the capacity not only to measure the proprioceptive posture but also to accommodate failure cases with sensor configuration redundancy. Furthermore, two soft finger modules can be seamlessly assembled into the two-segmented arm without redesign, integrating directly with the existing sensors to measure the states of finger grasping and releasing.

A. Proprioceptive Posture in Two-segmented Arm

Two SoRo-MIS linear actuators were parallelly arranged to

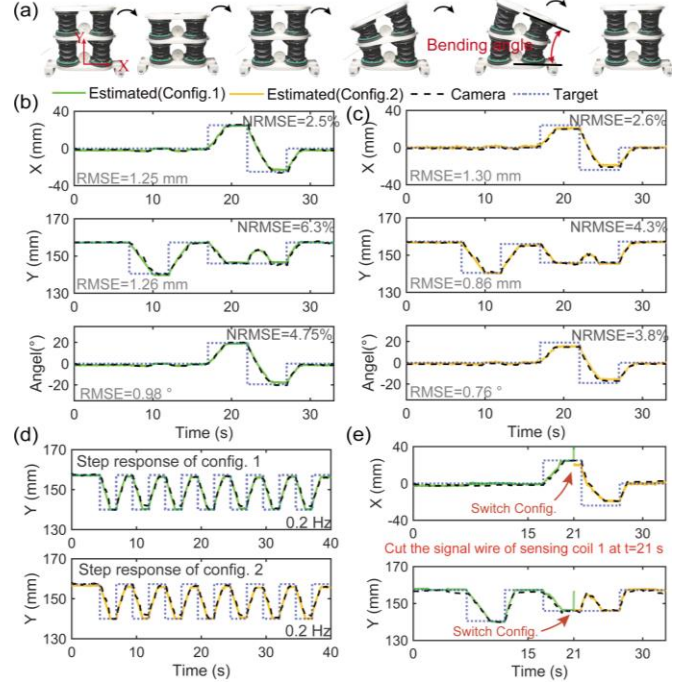


Fig. 7. Proprioceptive posture in a soft two-segmented robot for motion control. (a) Photos of motion control demonstrations based on two configurations. (b)(c) Proprioceptive posture sensing with Config. 1 and Config. 2. (d) Square wave function-following test of Config. 1 and Config. 2. (e) Real-time switch from Config. 1 to Config. 2.

one soft robotic bending module; two modules were stacked to construct a soft arm capable of working in a two-dimensional space, as shown in Fig. 6(a). We used a stereo camera (ZED Mini, 1.5% accuracy) to measure the mark at the center of the distal plate as the ground truth. The soft robot was equipped with a pneumatic system, which consists of eight solenoid valves (frequency: 30Hz max.) and pressure sensors (-100~300 kPa) providing positive and negative pressure control for four circuits with the aid of a microcontroller (STM32 F103). The air pressure was supplied by two micropumps (positive pressure: 140 kPa; negative pressure: -90 kPa), as shown in Fig. 6(b). With this configuration, a proprioceptive posture controller operating under the on-off modes of solenoid valves was implemented, as shown in Fig. 6(c). In this controller, the desired posture of the soft arm is converted into target voltages using Equation (10). These target voltages directly control the on-off states of the solenoid valves. To avoid overflow, a 1.5% voltage deadband was applied, balancing accuracy and actuation stability. This deadband ensures a control accuracy of ± 0.4 mm. The proposed soft robotic bending module is configured with one transmitting and two sensing coils for each, arranged with parallel center axes, resulting in a total of two transmitting and four sensing coils integrated into the arm. Their transmitting coils were separately supplied with $f = 22\text{kHz}, 24\text{kHz}; I_0 = 300\text{mA}$. Since the coil configuration differs from that of the linear actuators, fitting functions were recalibrated with $e_1 = 51.248, 42.466, e_2 = -0.2082, -0.2136$, and $e_3 = -42.723, -37.56$, separately for magnetic fields of frequencies 22kHz and 24kHz. These fitting functions apply to the soft robots designed with soft robotic bending modules.

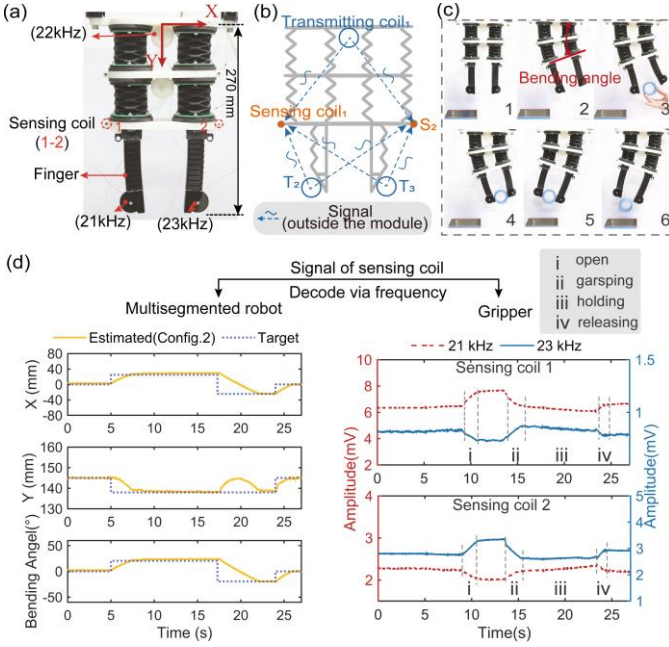


Fig. 8 Proprioceptive posture in a soft finger-assembled arm. (a) Prototype of the soft finger-assembled arm. (b) Signal transmission within the robot. (c) Photo of motion demonstration. (d) Arm position and finger motion information decoupled from composite signals of sensing coils.

In this robotic arm, two stacked modules overlap two magnetic fields, with two transmitting and four sensing coils forming a sensory array. This array enables flexible configurations for achieving the desired proprioceptive posture as shown in Fig. 6(d). Specifically, Configuration 1 collects signals from all sensing coils across both transmitting coils, while Configuration 2 collects signals from the two distal sensing coils of the bottom transmitting coil. This redundancy allows the system to not only measure proprioceptive posture but also enhance resilience to sensor failures.

Fig. 7(a)(b)(c) showcase the soft robot's capability to measure the proprioceptive state and track the spatial position using feedback control. Six target positions in the timeline for the endpoint were provided, including initial state (Target 1), contraction (Target 2), extension (Target 3), bending right (Target 4), bending left (Target 5), and returning to the initial state (Target 6). During the bending motion, the desired bending angles for both rightward and leftward bending of the arm were 19.5° . The sensing and control performance of the soft robot was demonstrated in both configurations, with Fig. 7(b) showcasing its performance in Configuration 1 and Fig. 7(c) showcasing its performance in Configuration 2. The proposed array achieved a total sensing range of approximately 150 mm. With Configuration 1, the RMSE in the X-axis and Y-axis directions were 1.25 mm and 1.26 mm, and the NRMSE was 2.5% and 6.3%, respectively. The RMSE of the bending angle was 0.98° , and the NRMSE was 4.75%. With Configuration 2, the RMSE in the X-axis and Y-axis directions was 1.3 mm and 0.86 mm, and the NRMSE was 2.6% and 4.3%, respectively. The RMSE of the bending angle was 0.76° , and the NRMSE was 3.8%. We conducted a 0.2Hz square wave function-following test in the Y direction to evaluate whether

different coil configurations would introduce control delays into the system, as shown in Fig. 7(d). The range in the Y direction is from 157mm to 140mm, corresponding to an amplitude of 17mm. The average step response time for Configuration 1 and Configuration 2 is 1.48s and 1.49s, respectively. The results indicate the same performance under different configurations.

To further demonstrate the potential of flexible configurations, we present a case of accommodating sensor failure. The proposed robotic arm followed the same six target positions under Configuration 1. To simulate sensor failure, we cut the wire of sensing coil 1 in the first segment at 21s, as shown in Fig. 7(e). The system detected the abnormal signal and automatically switched to Configuration 2, ensuring the continued operation of the arm. The results show that before 21s, when Configuration 1 provided sensory feedback, the arm moved from 140 mm to 157.3 mm along the Y-axis. The actual position reached 157.6 mm, resulting in a 0.3 mm deviation from the target. After 21s, Configuration 2 successfully maintained the target position with the same 0.3 mm deviation. This adaptability enhances the stability and reliability of the robot's sensory system.

B. Modular Design in Soft Finger-assembled Arm

The flexible coil configurations also enable a modular design that allows for the assembly and disassembly of different soft robotic modules without compromising the overall system or requiring a redesign. Two soft finger modules fabricated by 3D-printable thermoplastic polyurethane (TPU) were assembled into the distal plate of the two-segmented arm, constructed to a soft finger-assembled arm, as shown in Fig. 8(a). Utilizing two sensing coils at the distal plate of the two-segmented arm, each soft finger equipping a transmitting coil is enough for detecting the position changes of both fingers (Fig.8(b)). Therefore, the sensing coils at the distal plate of the arm receive signals not only from the transmitting coil on the bottom plate (22 kHz) but also from the transmitting coils on the soft fingers (21 kHz and 23 kHz). The soft finger-assembled arm successfully moved a cylindrical object from one point to another, involving coordinated control of both arm posture and finger states, as shown in Fig. 8(c). Fig. 8(d) illustrates the output signals of the sensing coils located on the left and right fingers. The arm was controlled to bend rightward by 20.5° to grasp the object, with a sensing bending angle of 21.8° . It then bent 20° to the left to release the object, with a sensing bending angle of 19.1° . Either sensing coil 1 or 2 can capture the monotonic signal, indicating the transition from the grasping to the holding state. It is demonstrated that integrating soft finger modules does not affect the measurement of arm posture, while simultaneously introducing new sensory information for detecting finger states. Furthermore, the incorporation of two transmitting coils into the fingers extends the overall sensing range from 160mm to 270mm.

C. Mutual Position Sensing in Two Crawling Robots

Two soft robotic bending modules were separately developed into two crawling robots (Robot A and B in Fig.

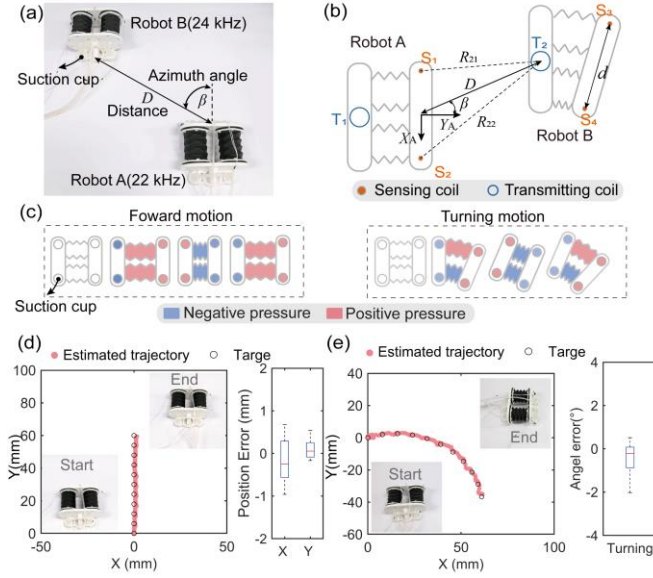


Fig. 9 Motion control of a single crawling robot. (a) Prototype of crawling robot. (b) Kinematics and mutual positioning of crawling robot (c) Motion gait of a crawling robot. (d) Controlled straight-line crawling of a crawling robot (e) Controlled curved-path crawling of a crawling robot.

9(a)), enabling the measurement of both proprioceptive posture and mutual positioning over a large sensing range. Each crawling robot performs both straight and turning movements through the linear and bending motions of the modules. To prevent slippage, four pneumatic suction cups are attached separately to the bottom of the head and tail plates. A total of two transmitting coils (Coil A with 22 kHz for Robot A and Coil B with 24 kHz for Robot B) and four sensing coils (Coil 1 and 2 for Robot A, and Coil 3 and 4 for Robot B) were involved in this robotic system, as shown in Fig. 9 (b). The arrangement of coils in crawling robots was the same as the soft robotic bending module, leading to identical parameters for the fitting functions. In this case, the magnetic field generated by the transmitting Coil B can not only be detected by Coil 3 and Coil 4 to measure the proprioceptive states of Robot B but also be recognized by Coil 1 and Coil 2 to determine the mutual positions between Robot A and Robot B. The mutual positions between the two robots were defined by the distance D and the azimuth angle β , which geometrically satisfy

$$\begin{cases} D = \sqrt{\frac{f_{R21}^2(E_{21}) + f_{R22}^2(E_{22})}{2} - \left(\frac{d}{2}\right)^2} \\ \beta = \arccos\left(\frac{(d/2)^2 + D^2 - f_{R21}^2(E_{21})}{dD}\right) \end{cases} \quad (12)$$

Fig. 9(c) illustrates the crawling gait during both forward and turning motions. In each gait cycle, we applied a 6 mm displacement or a 10° turning angle. The control system of a single crawling robot is identical to that of the two-segmented arm. This forward and turning gait, along with the ability to track its trajectory, was further validated. A coordinate system was established with the center of the crawling robot's head at the initial moment set as the origin. The resulting trajectory, shown in Fig. 9(d) and (e), demonstrates that the crawling robot can accurately follow both linear forward movements with a 1.6

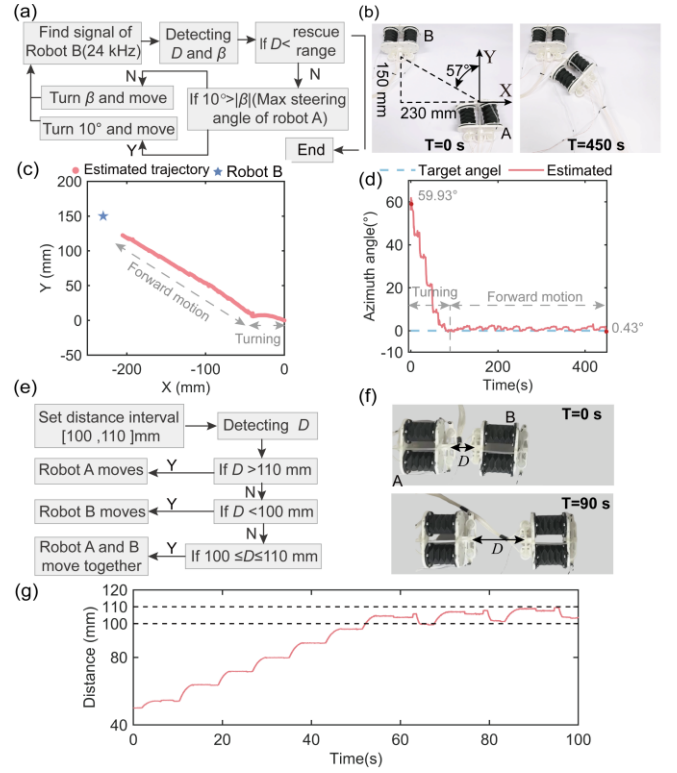


Fig. 10 Mutual position sensing in a two-crawling robot system for motion coordination. (a) Search strategy of Robot A for locating Robot B. (b) Photo of Robot A crawling towards Robot B. (c) Perception of Robot A's crawling trajectory. (d) Results of azimuth angle with time in the experiment. (e) Coordinated crawling strategy for Robots A and B with predefined distance maintenance. (f) Photo of coordinated crawling between Robot A and B. (g) Results of distance between the two robots with time in the experiment.

mm deviation over a 60 mm crawling displacement and turning motions under the proposed gait with a 3.5° deviation from the 100° target turning angle. During straight crawling, the median error in the X direction was -0.24 mm, with the 25th and 75th percentiles at -0.56 mm and 0.28 mm, respectively. The median error in the Y direction was 0.06 mm, with the 25th and 75th percentiles at -0.08 mm and 0.25 mm, respectively. During turning locomotion, the median angular error was -0.22° , with the 25th and 75th percentiles at -0.87° and 0.07° , respectively.

To verify the capability in both proprioceptive posture and mutual position sensing, we placed Robot A and Robot B in the environment, assuming that Robot B had lost mobility and was positioned at an unknown location. Benefiting from its ability to sense both proprioceptive posture and mutual positions, Robot A can search for Robot B by continuously adjusting its crawling gait, approaching within 20 mm in distance. This searching process is illustrated in Fig. 10(a), while Fig. 10(b) shows Robot A gradually crawling toward Robot B. A coordinate system was established at the head of robot A's initial position. On this coordinate system, it was observed in Fig. 10(c)(d) show that Robot A successfully located Robot B, adjusted its orientation accordingly, and ultimately approached it within the set distance. During the searching process, the motion of Robot A is divided into two phases: turning and forward crawling. In the first phase, the robot adjusts its head orientation, reducing the initial azimuth angle β from 59.93° toward 0° . In the second phase, the robot advances forward

while performing small corrective adjustments to keep the heading angle close to 0° , with a maximum deviation of 2.88° . Finally, Robot A stops at 37.5 mm from the center of Robot B's transmitting coil (coil radius: 15 mm), with a heading angle of 0.43° .

The capability of keeping a predefined distance between two moving crawling robots was further investigated. Robot B moved straight first, followed by Robot A. The control process of Robot A and Robot B is shown in Fig. 10(e). During this process, the distance between Robot A and Robot B was maintained within a range of 100 mm to 110 mm. Results in Fig. 10(f)(g) illustrate that the distance between two robots could always be maintained in the designated range, showcasing the system's ability to achieve mutual positioning during robot motion.

V. CONCLUSION

This study explores the utilization of a magnetic inductive sensing mechanism with multiple magnetic fields in soft pneumatic robots to track distributed key points, enabling flexible and versatile configurations of soft robotic modules over a large sensing range. We presented a Soft Robotic Magnetic Multifield Inductive Sensing (SoRo-MIS) approach with principles of defining proprioceptive posture of soft pneumatic robotic module through key distributed points and configuring magnetic coils into these points to extend the sensing range by overlapping multiple magnetic fields. We validated this concept by analyzing Signal-to-Noise Ratios (SNRs) over 20 dB under different coil locations and orientations, as well as by extending the sensing range in two linear actuators through the overlap of two magnetic fields at different frequencies. Each actuator achieves a linear displacement of over 20 mm, extending the sensing range beyond 40 mm with a sensing RMSE of 0.3 mm. This concept facilitated the modular design of soft robots, allowing for the assembly and disassembly of different soft pneumatic modules without compromising the overall system or requiring a redesign. It expanded the sensing functionality from proprioceptive posture to mutual positioning, showcasing the system's modularity and scalability. We modularly assembled two soft robotic bending modules into a two-segmented arm, showcasing the ability to measure proprioceptive posture with an RMSE of 1.25 mm over a 160 mm range while also ensuring resilience to sensor failure through flexible configurations. This two-segmented arm was also shown to seamlessly integrate soft finger modules without affecting arm posture measurement, while simultaneously introducing new sensory information for detecting finger states. This sensory information was further employed for target posture control. The two-segmented arm achieved the target position with a control accuracy of 0.3 mm. In addition to the target posture control, two soft bending modules were separately developed into two crawling robots, enabling the measurement of both proprioceptive posture and mutual positioning over a large sensing range. We demonstrated their capabilities in moving forward with a 1.6 mm deviation over a 60 mm crawling displacement and in

turning with a 3.5° deviation from the 100° target turning angle. Their crawling abilities further enhance potential applications in search scenarios involving interaction between two robots. One robot completed the tasks of searching for the other within a 200 mm space and following the moving robot while maintaining a 10 mm distance. Overall, the proposed study regarding sensing range, sensing accuracy, sensor size, simultaneous proprioceptive posture sensing, mutual position sensing, and modularity in soft robotic design is concluded in Table I.

Although the proposed sensing scheme demonstrates potential for modular assemblies, increasing the number of modules inevitably increases the number of coils, thereby imposing higher demands on hardware capacity (e.g., signal-source and DAQ channels). In addition, due to the required minimum frequency spacing between transmitting coils (1 kHz in this work), an excessive number of excitation frequencies may exceed the DAQ sampling limit and cause signal distortion. A potential solution is to employ a chip-select-based multiplexing strategy for coil excitation and sampling, reducing the instantaneous hardware load. Another limitation of this method is its vulnerability to electromagnetic noise interference and environmental temperature (Supplemental Materials), which may limit its applicability in certain situations. Future research will focus on developing strategies to mitigate interference and enhance the robustness of the approach. In addition, extending this framework to 3D motion and validating it experimentally is also a primary challenge that needs to be addressed in future work. Beyond addressing these limitations, the distinctive modularity and scalability of the proposed approach can benefit both sensor design and functional expansion. By employing flexible transmitting coils, external interaction forces induce coil deformation and corresponding magnetic field variations that can be detected by the sensing coils. Furthermore, by combining multiple modalities of sensing information, this approach shows potential for expandable sensing functions, not only in sensing modalities but also in spatial scale.

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